



Comparative performance analysis of machine learning algorithms on CICIDS2017 dataset for Intrusion detection

Group 10

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What is an Intrusion Detection System (IDS)?

- ▶ A security mechanism that monitors network activity.



Why IDS Matters / Applications

- ▶ Detects cyberattacks.
- ▶ Identifies Zero-Day or unknown threats.
- ▶ Automates monitoring for massive traffic volumes.

Traditional vs. ML Based IDS

Feature	Signature-Based (Traditional)	Machine Learning-Based
Detection	Known attack patterns	Unknown/Anomalous behavior
Threat Type	Strong for known threats	Better for evolving threats
Updates	Needs manual rule updates	Learns from data patterns
Zero-Day	Limited capability	Highly adaptive

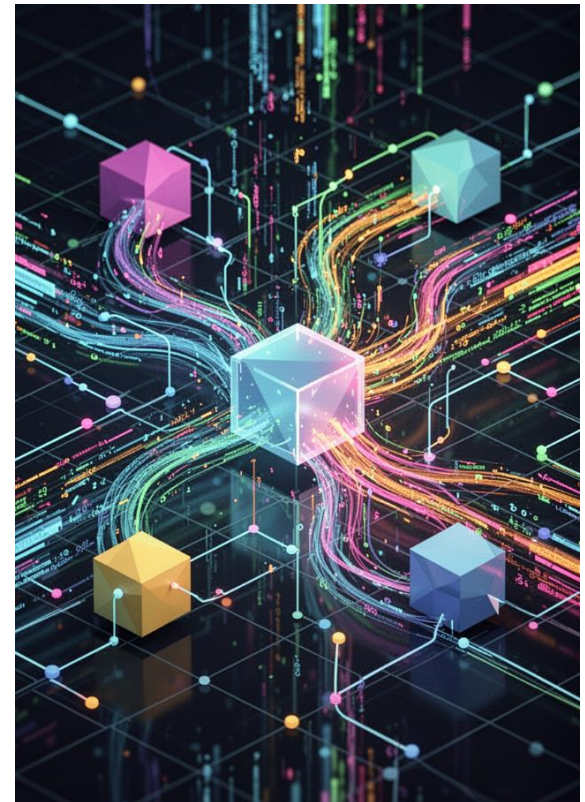


Research Question

Which category of ML algorithms is fundamentally better suited for IDS?

Research Gap

- ▶ Existing studies focus mostly on raw accuracy.
- ▶ Lack of explanation on 'why' specific classes perform better.
- ▶ Limited work on grouping algorithms by learning principles.
- ▶ Need for a common pipeline comparison across families.



Why Not Just Tune One Model?

- ▶ Hyperparameter tuning (depth, rate) offers incremental gains.
- ▶ Tuning doesn't answer in depth questions like:
 - ▶ Are Ensemble methods inherently better?
 - ▶ Do Neural models justify their high complexity?

Our Proposed Work

- ▶ Category-based empirical study using CICIDS2017.
- ▶ Compare multiple Machine Learning Categories
- ▶ Objective: Identify which categories are most robust and why.

Methodology

- ▶ Data preprocessing
- ▶ Feature selection
- ▶ Handling class imbalance
- ▶ Train
- ▶ Evaluate
- ▶ Analyze

Proposed Categories & Algorithms

Category	Example Algorithms
Linear Models	Logistic Regression, SGD Classifier
Probabilistic	GaussianNB, MultinomialNB
Anomaly Detection	Isolation Forest, One Class SVM
Tree-Based	Random Forest, XGB Classifier
Neural Models	MLP, Auto Encoder

How We Measure Performance

- ▶ Accuracy
- ▶ Precision
- ▶ Recall.
- ▶ F1-Score (Crucial for imbalanced data).

Benchmark Cases

- ▶ Toward Generating a New Intrusion Detection Dataset and Intrusion Traffic Characterization [2019] [[source](#)]
- ▶ Comparative Performance Evaluation of Machine Learning Algorithms for Cyber Intrusion Detection (2024) [[Source](#)]
- ▶ Robust Anomaly Detection in Network Traffic: Evaluating Machine Learning Models on CICIDS2017 [2025] [[source](#)]

Expected Contributions

- ▶ Provide a structured category-based comparison.
- ▶ Identify learning principles best suited for IDS traffic.
- ▶ Help researchers choose models rationally based on data behavior.
- ▶ Support practical IDS design decisions.

Available Datasets and Limitations

- ▶ **DARPA 98/99** — Synthetic Bias
- ▶ **KDD Cup 99** — Extreme Redundancy
- ▶ **NSL-KDD (2009)** — Outdated
- ▶ **UNSW-NB15 (2015)** — Lacks complexity of traffic
- ▶ **CICIDS2017 (2017)** — Class Imbalance

Selected Dataset : CICIDS2017

Why ?

- ▶ 2.8 Million Records
- ▶ Complete and Realistic setup with different devices and OS
- ▶ 80 Network flow features
- ▶ Both Benign traffic and common attacks (14 Attack Types)
- ▶ No anonymization of data.

Conclusion

- ▶ ML improves IDS by detecting complex threats.
- ▶ Moving beyond individual model comparison to category analysis.
- ▶ CICIDS2017 provides the realistic data needed for validation.
- ▶ Goal: From performance scores to meaningful understanding.

Experimental Results & Analysis

Model	Precision	Recall	F1 Score	Balanced Accuracy	MCC	ROC-AUC	PR-AUC	Specificity	G-Mean
Random Forest	0.998181	0.998210	0.998190	0.838114	0.994041	0.996112	0.895284	0.999606	0.998907
XGBoost Classifier	0.998433	0.998589	0.998439	0.807852	0.995305	0.999937	0.930121	0.999668	0.999129
MLP+	0.761200	0.627600	0.654000	0.627600	0.976300	0.978300	0.669800	0.997800	0.680400
Autoencoder + MLP	0.972000	0.973900	0.972300	0.497200	0.910900	0.996400	0.994000	0.900300	0.936400
Isolation Forest	0.499940	0.295110	0.371140	0.617563	0.294037	0.753493	0.369515	0.940016	0.526695
Fast One-Class SVM	0.377528	0.712031	0.493432	0.736730	0.380729	0.814041	0.628104	0.761429	0.736316
Logistic Regression	0.951815	0.775140	0.841156	0.893757	0.597053	0.978701	0.978737	0.984761	0.873686
SGD Classifier	0.952766	0.876160	0.910261	0.686201	0.686201	0.966010	0.967408	0.986359	0.929628
Multinomial Naive Bayes	0.849900	0.887900	0.865200	0.163700	0.581000	0.904300	0.220000	0.972600	0.736400
Gaussian Naive Bayes	0.909200	0.173800	0.200600	0.585800	0.219800	0.834300	0.874000	0.072500	0.269200

